MINI PROJECT PRESENTATION
RESEARCH PROJECT MANAGEMENT SYSTEM

PROJEXPERT

Department of Computer Science and Engineering

Where Research Meets Efficiency

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Under the guidance of
Dr Tribikram Pradhan

Outline of the Presentation

- Introduction
- Initial System Study
- Feasibility Analysis
- Software Requirement Specification
- Schematic Diagram
- Data Flow Diagram
- Implementation
- Future Refinements
- Conclusion
- Motivation

Introduction

Research Project Management System aims to streamline the complex process of managing research projects by integrating all essential functions into a single platform. The system provides a comprehensive solution by enabling users, who are primarily researchers, to register with unique IDs, form research groups, and propose project topics under the supervision of selected staff members.

The scope of **ProjExpert** encompasses the entire lifecycle of a research project, from initial registration and group formation to final evaluation and feedback. It is designed to be used by academic and research institutions, aiming to improve overall efficiency and productivity in research project management.

Initial System Study

- Data Collection and Storage
  Researchers and project managers collect data manually, often recording information in paper files or basic digital documents. Financial records, project milestones, and team communications are stored in separate files, making centralised access difficult.

- Project Tracking and Coordination:
  Spreadsheets are commonly used to track project progress, deadlines, and budgets. Team members communicate via email or standalone messaging apps, leading to fragmented communication streams.

- Documentation and Reporting:
  Reports and documentation are created manually, often leading to inconsistencies and errors. Access to historical data is cumbersome, as it requires manual searches through various files and emails.

- Security and Access Control:
  Sensitive project data is stored in unencrypted files or within unsecured email communications. Access control is minimal, often relying on password-protected files or folders, which can be easily bypassed or mismanaged.

Feasibility Analysis

- Technical Feasibility

JavaScript

mongoDB

MongoDB Atlas

mongoDB

socket.io

- Operational Feasibility

- Economic Feasibility

Software Requirement Specification

FUNCTIONAL REQUIREMENTS

1. Admin Functionality:

Empower administrators to manage panel appointments, view all topics, and oversee the evaluation process.
Provide administrators with the ability to send notices to staff or users, attaching relevant documents as needed.

2. User Management:

Allow users to register with unique IDs and roles (researcher, supervisor, admin) using email and password.
Enable user authentication for secure access to the platform

3. Staff Management:

Provide supervisors with access to staff details, allowing them to assign co-supervisors and manage comments. Allow supervisors to view notices from the administrator and respond accordingly.

4. Group Formation:

Provide functionality for researchers to create user groups, appoint leaders, and add members with unique IDs.
Allow users to define project topics and select supervisors for their research projects within the group formation process.

5. Supervisor Interaction:

Implement a chat feature for seamless communication between researchers and their assigned supervisors.
Enable researchers to submit project-related documents and receive feedback from supervisors.

6. Document Management:

Allow researchers to submit project-related documents and repositories for evaluation by supervisors and panel members.

7. Panel Evaluation:

Designate staff members as panel members to evaluate research topics and projects submitted by researchers.
Enable panel members to provide feedback and assessments on the submitted projects.

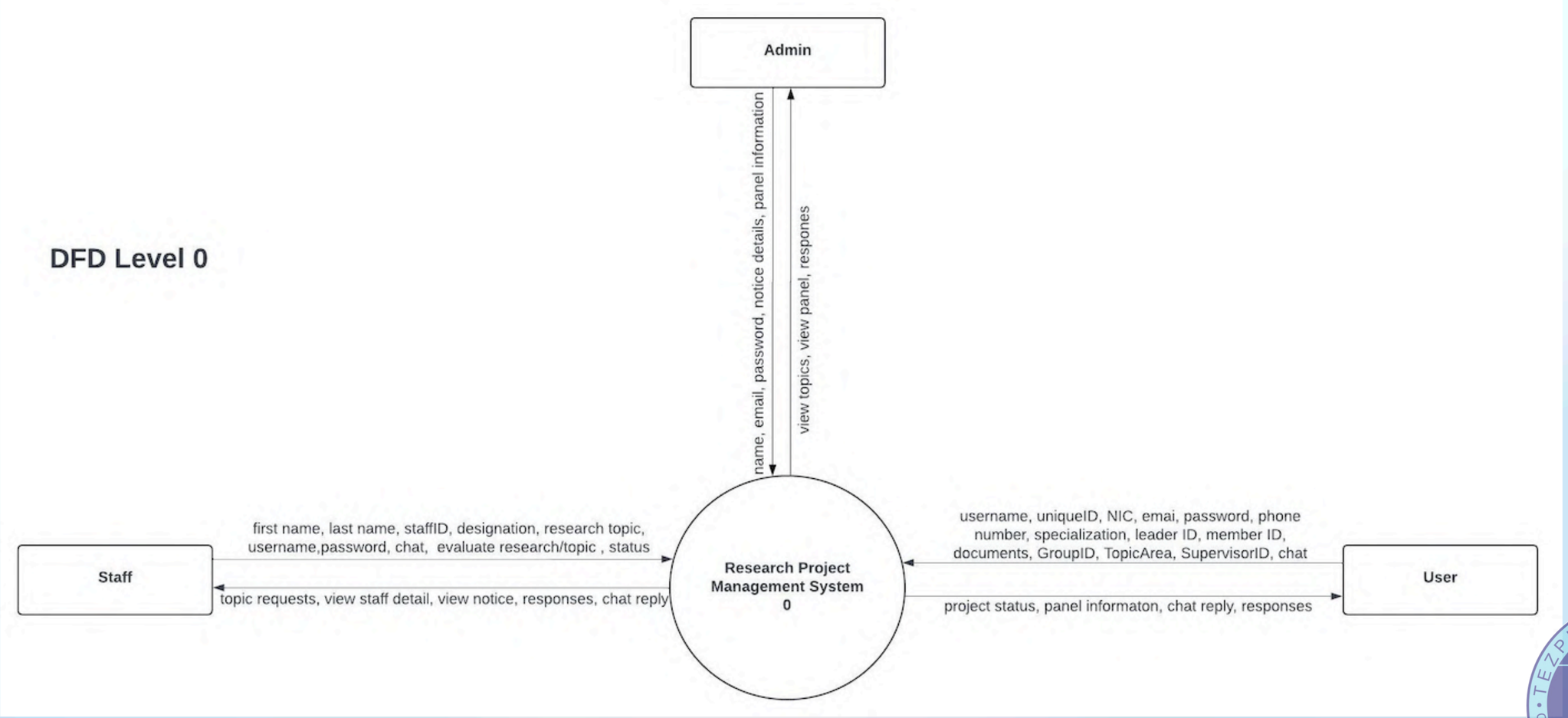
8. Project Tracking:

Enable researchers to track the progress of their projects, including the status of evaluations and the assigned panel members.
Provide real-time updates on project milestones and deadlines to ensure timely completion.

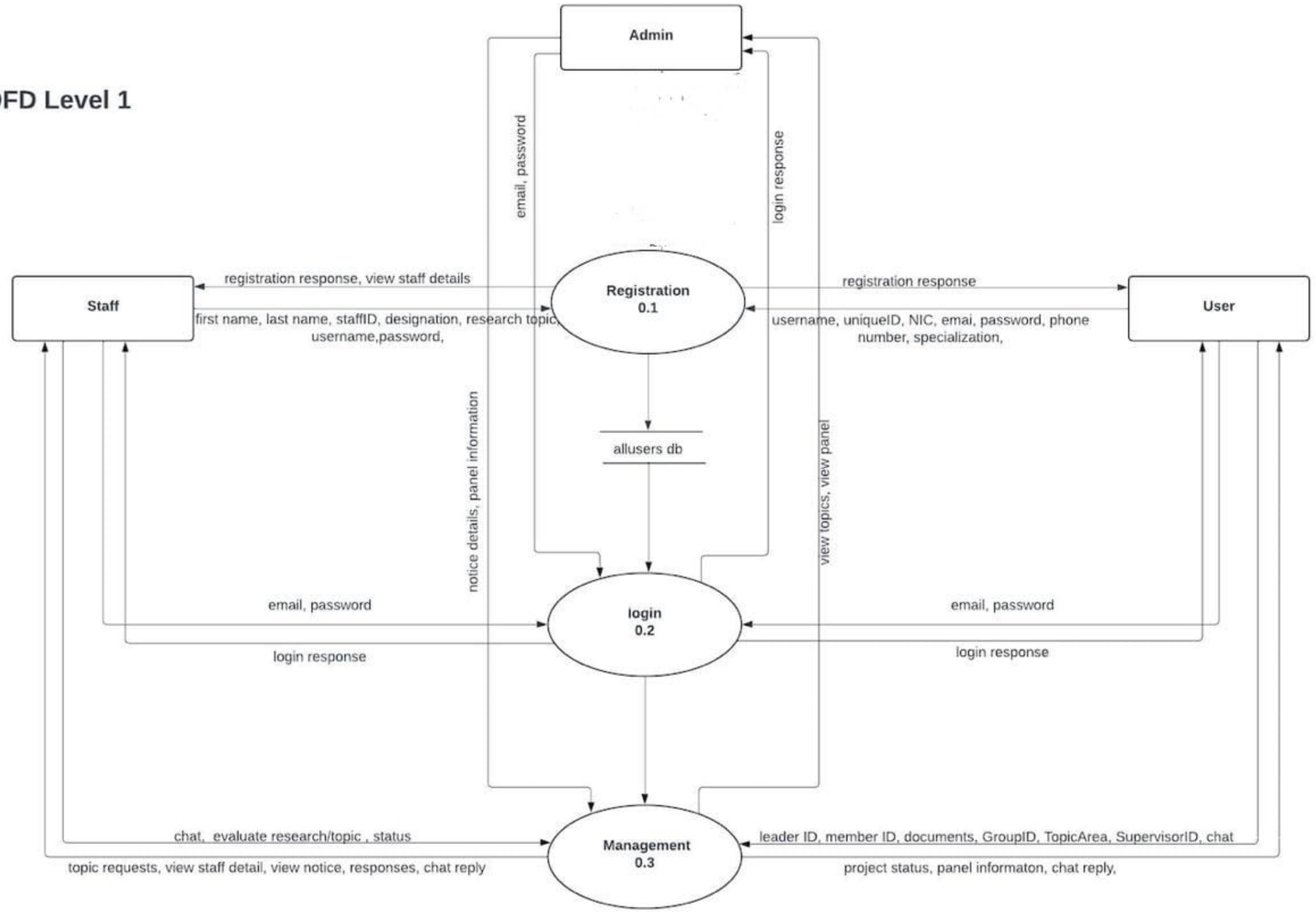
NON FUNCTIONAL REQUIREMENTS

Schematic Diagram

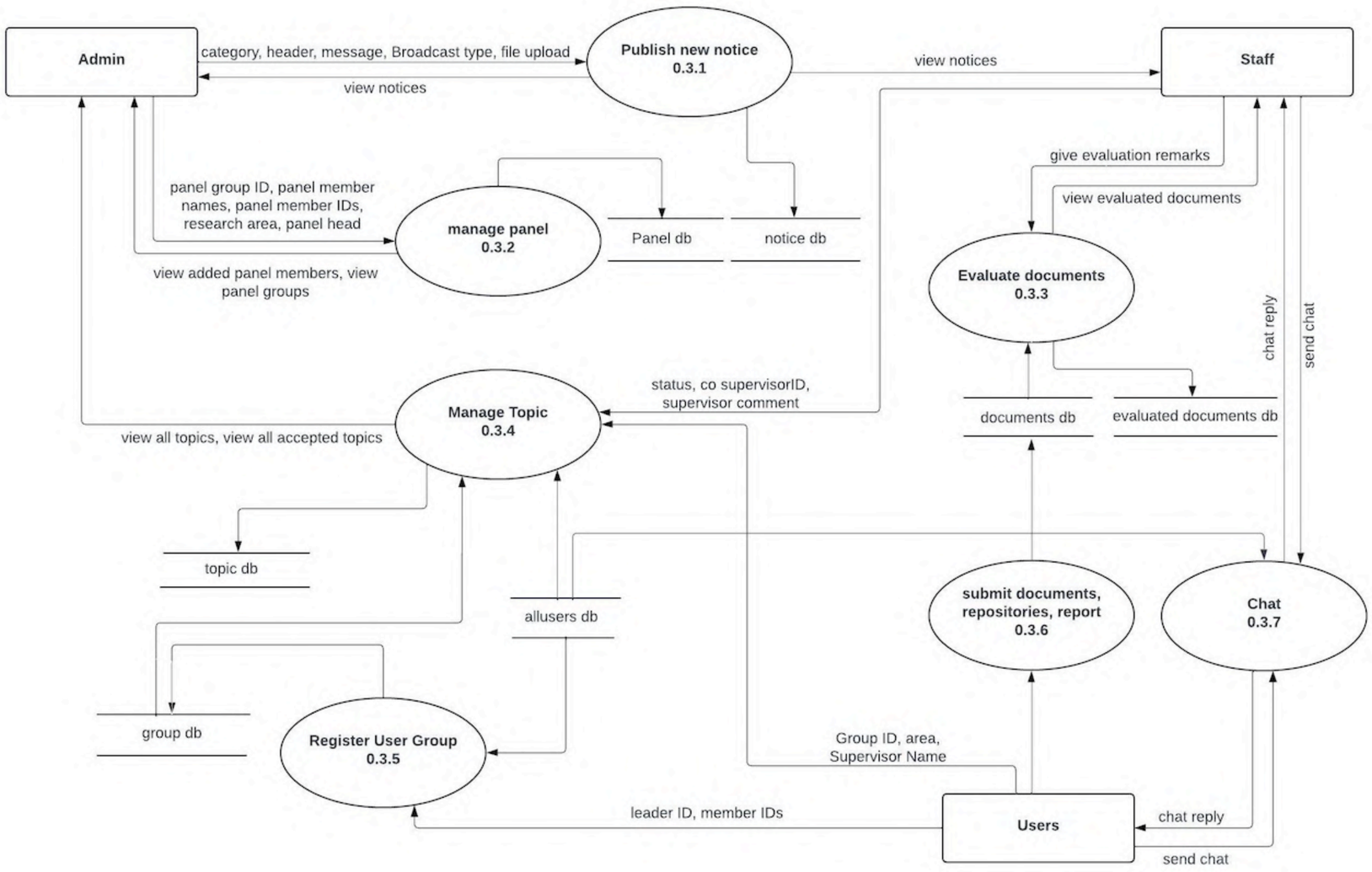
Data Flow Diagram



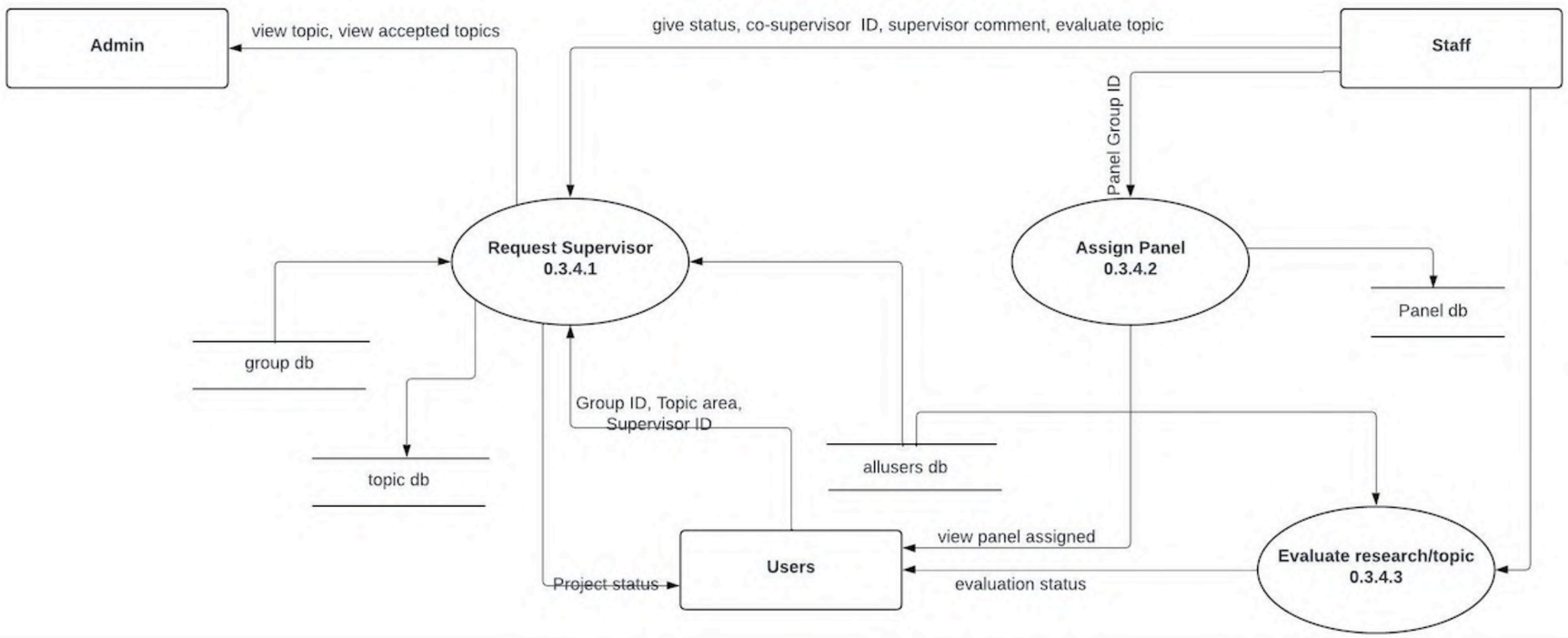
DFD Level 1


Admin
email, password
login response
Staff
registration response, view staff details
first name, last name, staffID, designation, research topic, username,password,
Registration 0.1
registration response
username, uniqueID, NIC, emai, password, phone number, specialization,
User
allusers db
notice details, panel information
view topics, view panel
email, password
login response
login 0.2
email, password
login response
chat, evaluate research/topic , status
topic requests, view staff detail, view notice, responses, chat reply
Management 0.3
leader ID, member ID, documents, GroupID, TopicArea, SupervisorID, chat
project status, panel informaton, chat reply,

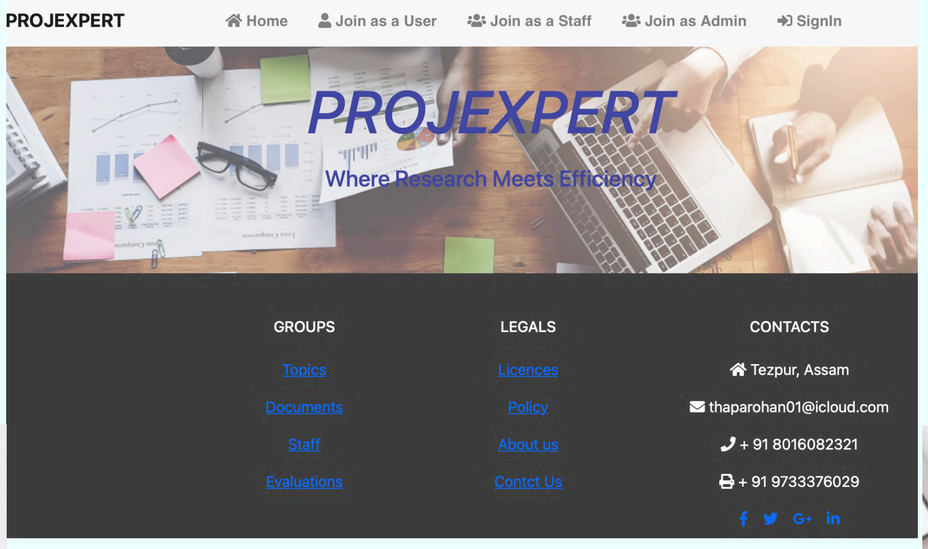
DFD Level 2

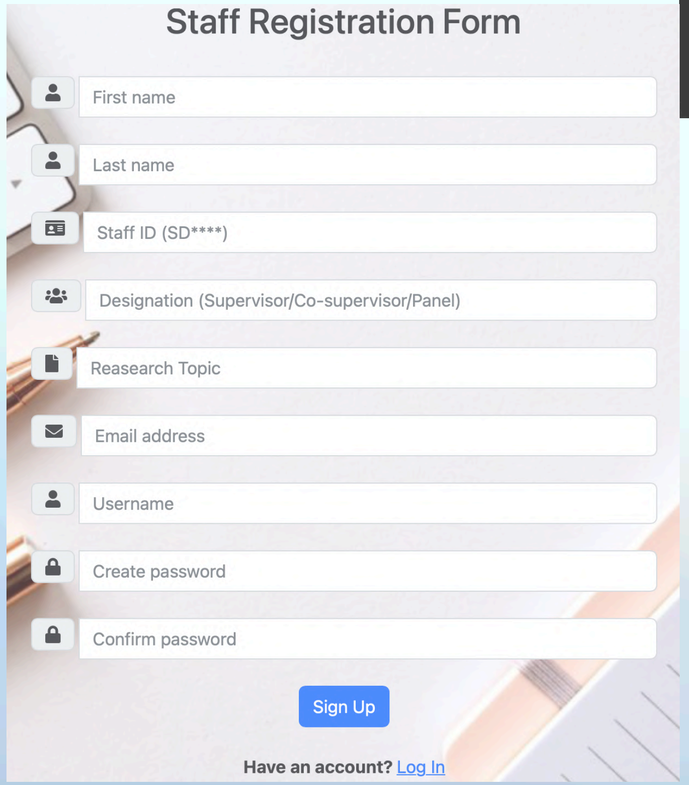

Admin
category, header, message, Broadcast type, file upload
view notices
Publish new notice
0.3.1
view notices
Staff
give evaluation remarks
view evaluated documents
panel group ID, panel member names, panel member IDs, research area, panel head
manage panel
0.3.2
Panel db
notice db
view added panel members, view panel groups
Evaluate documents
0.3.3
chat reply
send chat
status, co supervisorID, supervisor comment
Manage Topic
0.3.4
documents db
evaluated documents db
view all topics, view all accepted topics
topic db
allusers db
submit documents, repositories, report
0.3.6
Chat
0.3.7
group db
Register User Group
0.3.5
Group ID, area, Supervisor Name
leader ID, member IDs
Users
chat reply
send chat

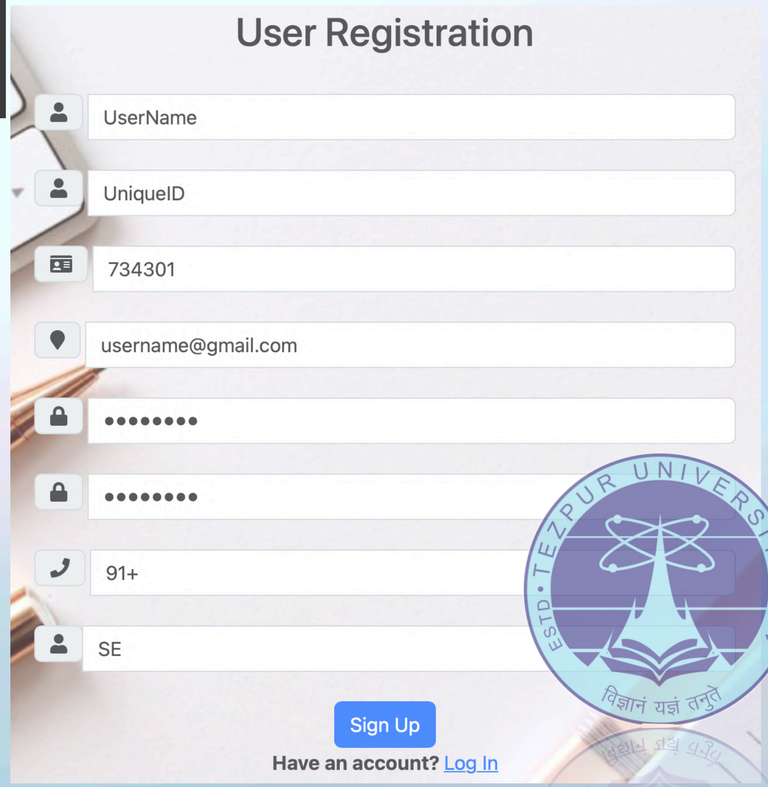
DFD Level 3


Admin
view topic, view accepted topics
give status, co-supervisor ID, supervisor comment, evaluate topic
Staff
Panel Group ID
Request Supervisor
0.3.4.1
Assign Panel
0.3.4.2
Panel db
group db
Group ID, Topic area,
Supervisor ID
topic db
allusers db
view panel assigned
Users
Project status
evaluation status
Evaluate research/topic
0.3.4.3

Implementation







ADMIN DASHBOARD

Notice Management | Topic Management | Panel Management

MANAGE NOTICES

Publish New Notice | View All Notices

Create Notice

Notice Category (topic/notice/presentation/document)

notice

Notice Header

Panel Assigned

Message

Panel Head Nabajyoti Medhi Assigned

Broadcast Type (all/supervisor/co-supervisor/panel)

all

Upload Template / Material

GRP834.pdf

Choose file No file chosen

Publish

All Notices

Count	Notice Category	Broadcast Type	Notice Header	Message	Downloadable Content	Action

TOPIC MANAGEMENT

All Topics | Accepted Topic

All Topics

Count	Topic	Group ID	Supervisor ID	Co-Supervisor ID	Status	Remarks

All Accepted Topics

Count	Topic	Group ID	Supervisor ID	Co-Supervisor ID	Documents	Assign Panel	Panel ID	Status	Comment

MANAGE PANELS

Add Panel Members | Create Panel Groups | View All Panels

ADD PANEL MEMBER

Panel ID

Enter Panel Group ID

Panel Member IDs

Enter Panel Member IDs

Panel Member Names

Enter Panel Member Names

Research Area

Enter Research Area

Save

ADD PANEL

Panel Group ID

Enter Panel Group ID

Panel Head

Enter Panel Head

Panel Research Area

Enter Panel Research Area

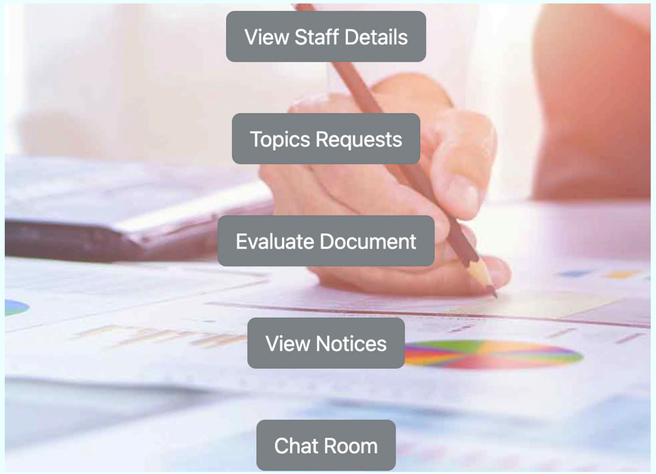
SAVE

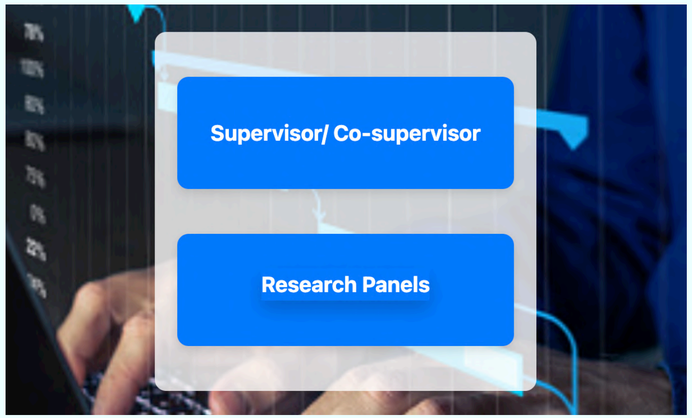
Panel Details

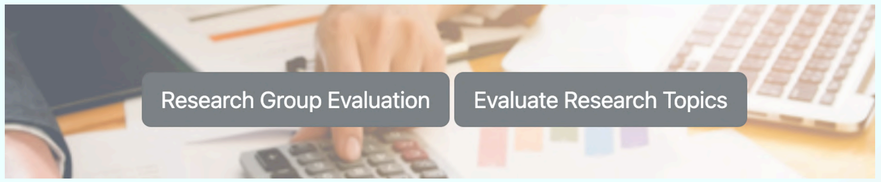
Search ...

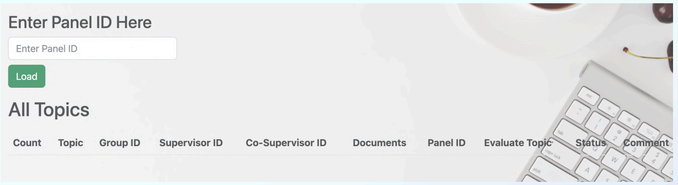
Index	panel Group ID	panel Head ID	Research Area

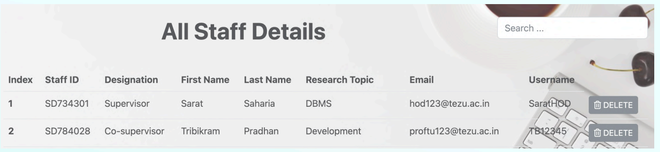
Implementation

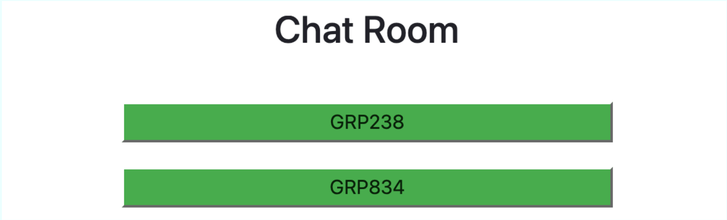


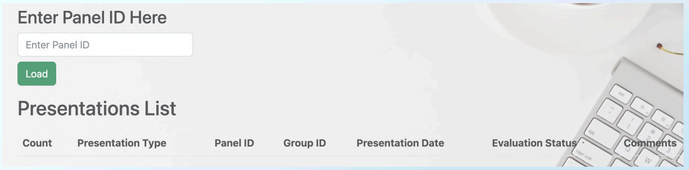


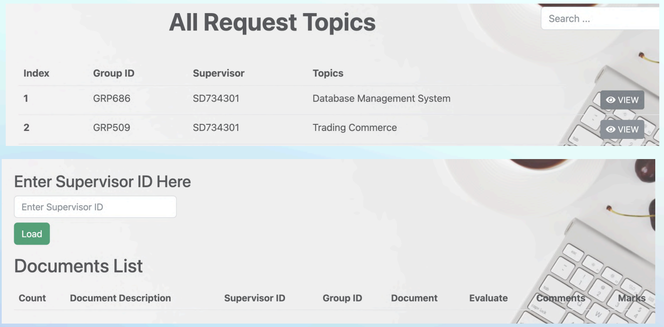


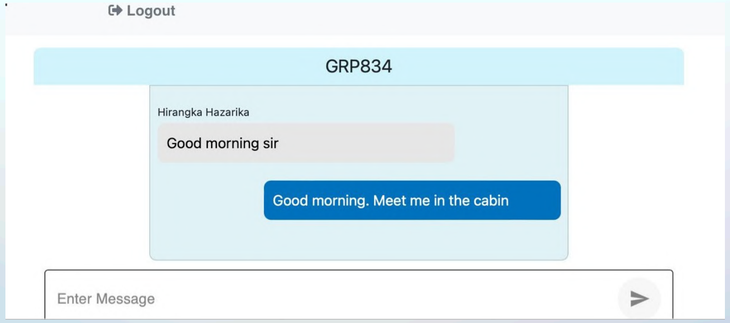


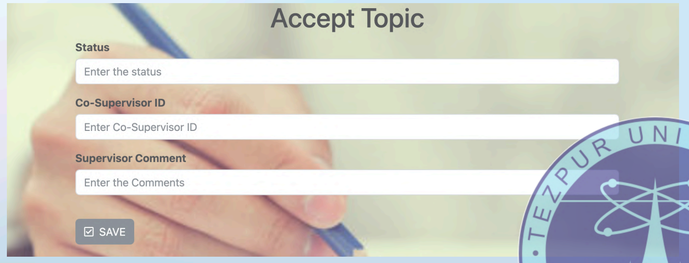


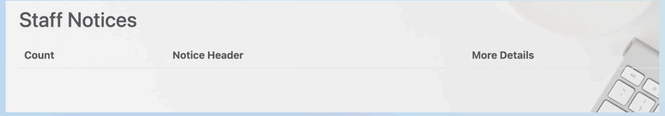




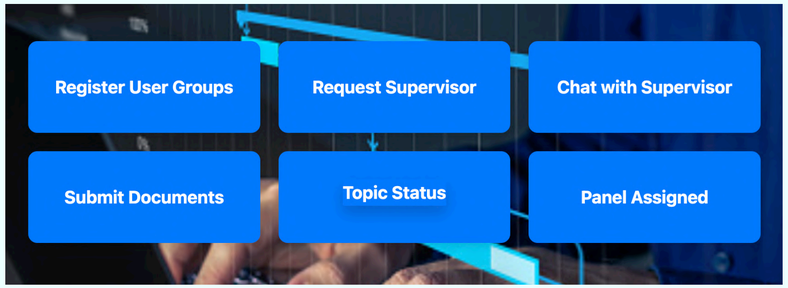


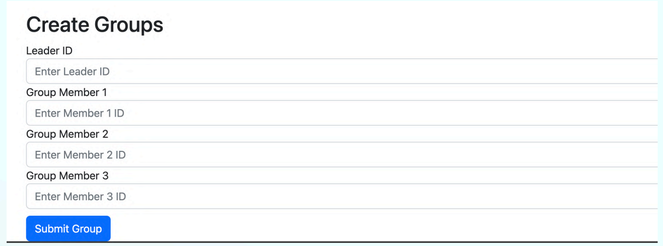


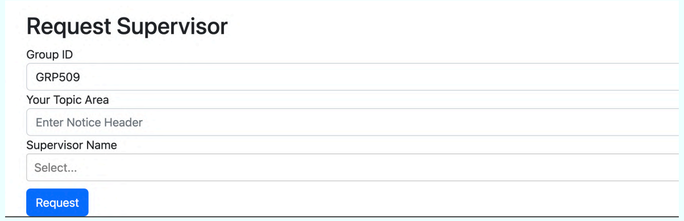


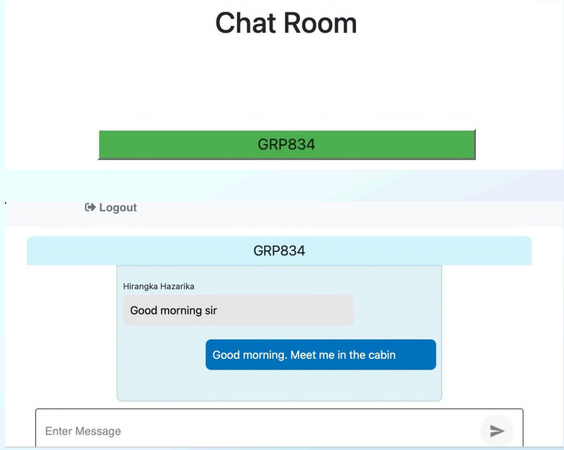


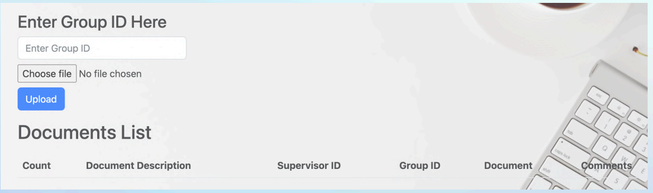
Implementation

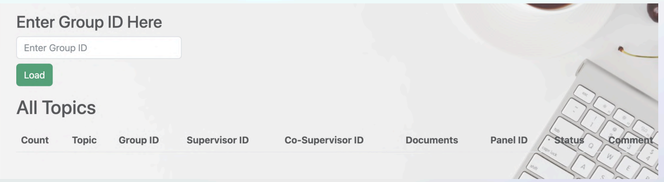


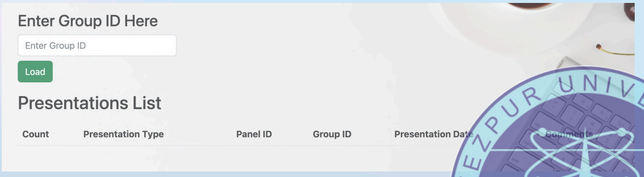




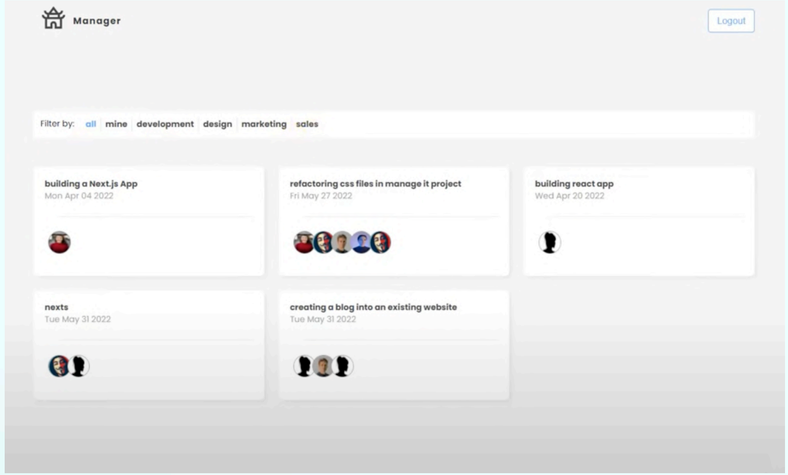


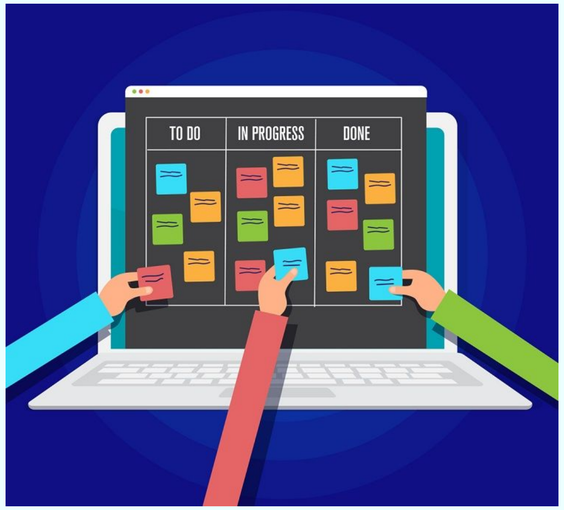


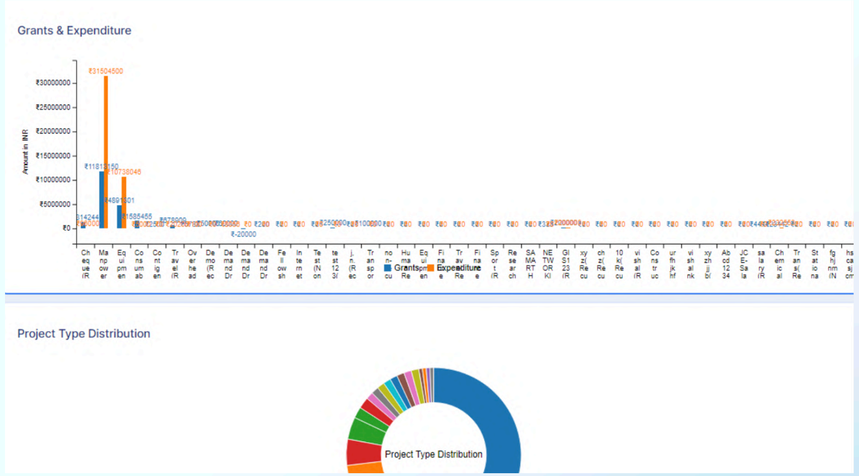




Future Refinements







Conclusion

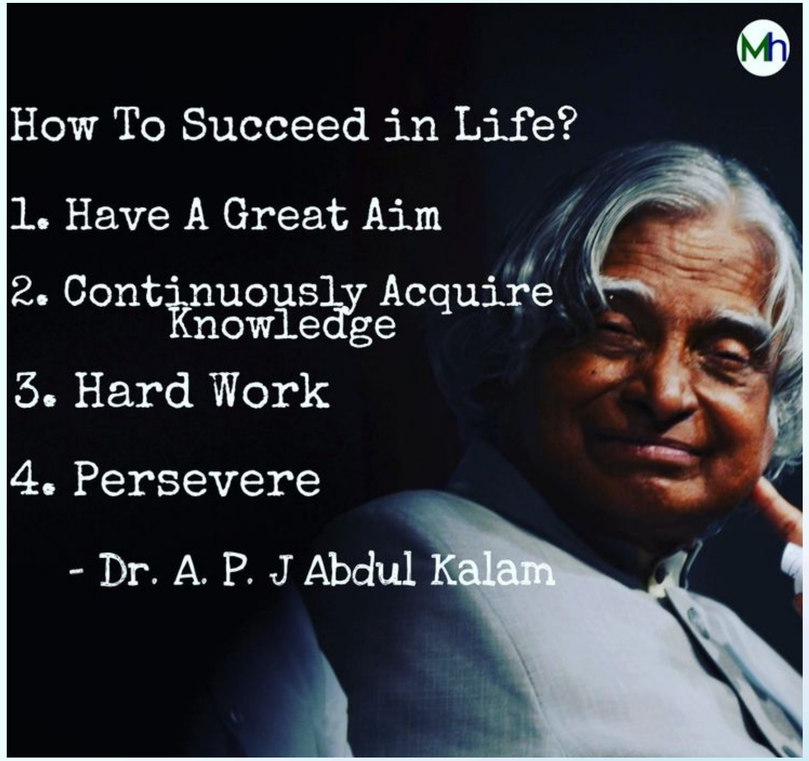
ProjExpert, a Research Project Management System represents a transformative solution for research project management, offering a comprehensive platform designed to streamline workflows, enhance collaboration, and drive innovation. By leveraging advanced technologies and user-centric design principles, ProjExpert empowers researchers, supervisors, and staff to efficiently manage projects, communicate effectively, and achieve their goals. With its intuitive interface, robust features, and scalability, ProjExpert can revolutionize the way research projects are conducted, ultimately contributing to advancements in knowledge and breakthroughs in various fields.

Despite its current focus on essential communication and document management features, there is significant potential for future enhancements for more collaborative features

Motivation

I would take this opportunity to express my gratitude to all the Professors and Faculty Members in the department for their teachings and academic support and being instrumental in the successful completion of this project.

I want to thank my guide for his professional guidance, unwavering support, patience, and inspiration.



THANK YOU